

Grade 7 Mathematics — Mock Exam — Paper D

HARD • NON-CALCULATOR • 75 MINUTES • 80 MARKS

NAME

DATE

SCORE / 80

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to work without one.
- **75 minutes for 80 marks** — about a minute a mark. If a 3-mark question has taken you five minutes, leave it and come back to it.
- No formulas and no methods are given anywhere on this paper — not one. Remembering the right one is part of what is being tested.
- Give every fraction and every ratio in its simplest form, and convert to a common unit before you simplify.
- The mark for each question is shown in square brackets on the right.
- Nothing tells you which topic a question belongs to, and several need two topics at once. Work out what you are being asked before you start writing.

SECTION 1 – QUICK FIRE

1. Write down the answer to each of these.

(a) How many prime numbers are there between 20 and 30? [1]

(b) $-56 \div -7$ [1]

(c) $\sqrt[3]{343}$ [1]

(d) Write $9^2 \times 9^5$ as a single power of 9. [1]

(e) $0.5 \div 0.01$ [1]

2. Write down the answer to each of these.

(a) Round **7.0962** to 2 decimal places. [1]

(b) 1.2×0.03 [1]

(c) Write $\frac{4}{15}$ as a decimal. Use dot notation. [1]

(d) Write the ratio 40 minutes : 2 hours in its simplest form. [1]

(e) Factorise fully $20a^2 + 15a$. [1]

SECTION 2 – SHORT ANSWERS

3. Two numbers are 18 and 45.

(a) Find the highest common factor (HCF) of 18 and 45. [1]

(b) Find the lowest common multiple (LCM) of 18 and 45. [1]

4. Simplify $(5^4 \times 5^3) \div 5^5$. Give your answer as an ordinary number. [2]

5. Work out:

(a) 62×0.01

[1]

(b) $0.062 \div 0.01$

[1]

6. Write these decimals in ascending order — smallest first.

[2]

0.303 0.033 0.33 0.3003

7. Write these fractions in descending order — largest first. You must show your working.

[2]

$\frac{7}{8}$ $\frac{5}{6}$ $\frac{11}{12}$ $\frac{17}{24}$

8. Work out $7\frac{2}{5} - 3\frac{5}{8}$. Give your answer as a mixed number in its simplest form.

[2]

9. A taxi charges a fixed fee of ₹50 plus ₹18 for each kilometre travelled.

(a) Write an expression for the cost of a journey of k kilometres.

[1]

(b) Write an expression, in its simplest form, for the cost of two such journeys.

[1]

10. Expand and simplify $2(5x - 3) - 5(x - 2)$. [2]

SECTION 3 – MULTI-STEP

11. Work out:

(a) -9×-6 [1]

(b) $-9 - 6$ [1]

(c) $-144 \div 12$ [1]

12. Work out $2.622 \div 0.038$, showing your method. [3]

13. Work out $14 \times 1\frac{5}{7}$. [3]

14. Work out $10 \div \frac{2}{7}$. [3]

15. Temperatures in degrees Celsius (C) and degrees Fahrenheit (F) are connected by $F = 1.8C + 32$.

(a) Find F when $C = 25$. [1]

(b) Find C when $F = 5$. [2]

16. Expand and simplify $7(2x - 3) - 4(2x - 6)$. [3]

17. Solve $4(2x - 3) = 3(x + 6)$. Show every step. [3]

- 18.** A car is bought for ₹640 000. One year later it is worth ₹544 000. Work out the percentage decrease in its value. [3]

SECTION 4 – PROBLEMS

- 19.** ₹7200 is shared between Asha and Bina in the ratio 5 : 4.

(a) Work out how much each of them receives. [2]

(b) Asha then gives 15% of her share to Bina. Work out how much she gives. [1]

(c) Work out how much each of them has now. [2]

Your answer to (c) should still add up to 7200. If it does not, one of the two amounts is wrong.

- 20.** A rectangle has a perimeter of 58 cm. Its length is 5 cm more than twice its width. The width is w cm.

(a) Form an equation in w . [2]

(b) Solve your equation to find the width. [2]

(c) Write down the length of the rectangle. [1]

21. A machine fills bottles that each hold 0.35 litres. In one run it uses 91 litres of juice.

(a) Work out how many bottles it fills in one run. [3]

(b) The juice costs ₹8 per litre. Work out the cost of the juice in one bottle. [2]

22. $n = 2^2 \times 3 \times 5^2$ $m = 2^3 \times 3^2 \times 5$

(a) Find the highest common factor (HCF) of n and m . [2]

(b) Find the lowest common multiple (LCM) of n and m . [2]

(c) Write n as an ordinary number. [1]

You do not need to work out n and m first. Compare the indices.

23. A laptop costs ₹45 000. In a sale its price is reduced by 20%. After the sale the reduced price is increased by 25%.

(a) Work out the sale price. [2]

(b) Work out the price after the increase. [2]

(c) Write down the overall percentage change from ₹45 000. [1]

24. Rahul has handed in this homework. Both answers are wrong. For each one, say what he did wrong and then write the correct answer.

(a) "The counters are red : blue = 3 : 5, so $\frac{3}{5}$ of the counters are red." [2]

(b) " $4 - 2(x - 5) = 4 - 2x - 10 = -2x - 6$ " [3]

A mark for spotting the mistake, a mark for fixing it. Naming the rule he broke is worth more than writing the right number.

End of paper. Go back to the two questions you were least sure of. On a mock, the marks you find in the last five minutes are the cheapest ones on the paper.